COL703/7203

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$\mathcal{A} \models \phi$

A first order formula ϕ over a signature σ is satisfiable if $\mathcal{A} \models \phi$ for some σ-structure $\mathcal{A}$.

If ϕ is not satisfiable then ϕ is called unsatisfiable.

If $\neg\phi$ is satisfiable then ϕ is called falsifiable.

If $\neg\phi$ is not satisfiable then ϕ is called valid.

The satisfiability and validity problems for FOL are undecidable.

Validity is semi-decidable

Satisfiability is undecidable



First order logic is sound and complete.

Relevance Lemma

Let A and A' be σ-assignments with the same universe and the same interpretation for constant, function symbols, predicate symbols in σ.

If A and A' give the same interpretation to every variable occurring free in a formula ϕ

then $A \models \phi$ iff $A' \models \phi$.

Examples

$\forall x \forall y \quad P(x,y) \rightarrow P(y,x)$

$\forall x \exists y \quad P(x,y)$ || falsifiable

$\exists x \exists y \left(P(x) \wedge \neg P(y) \right)$

$\forall x \; P(x) \rightarrow P(a)$ valid

$\forall x \left(P(x) \wedge Q(x) \right) \leftrightarrow \forall x \, P(x) \wedge \forall x \, Q(x)$

$$\forall x \big(P(x) \to Q(x) \big) \to \big(\forall x\, P(x) \to \forall x\, Q(x) \big)$$

valid (exercise: prove this)

$$\big(\forall x\, P(x) \to \forall x\, Q(x) \big) \to \forall x \big(P(x) \to Q(x) \big)$$

falsifiable

Universe $= \{a, b\}$

$P = \{a\}$ $Q = \{b\}$

Logical equivalence

ϕ and ψ are logically equivalent ($\phi \equiv \psi$) iff for all σ-assignments A

$$A \models \phi \quad \text{iff} \quad A \models \psi.$$

$$\neg \forall x \phi \equiv \exists x \neg \phi$$

$A \models \neg \forall x \phi$ iff $A \not\models \forall x \phi$

iff $A_{[x \mapsto a]} \not\models \phi$ for some $a \in U_A$

iff $A_{[x \mapsto a]} \models \neg \phi$ for some $a \in U_A$

iff $A \models \exists x \neg \phi$

$\forall x \phi \wedge \psi \equiv \forall x (\phi \wedge \psi)$

assuming that x is not free in ψ.

$A \models \forall x \phi \wedge \psi$ iff $A \models \forall x \phi$ and $A \models \psi$

iff for every $a \in U_A$ $A_{[x \mapsto a]} \models \phi$ and $A \models \psi$

iff ..... and $A_{[x \mapsto a]} \models \psi$

iff for every $a \in U_A$ $A_{[x \mapsto a]} \models \phi \wedge \psi$

iff $A \models \forall x (\phi \wedge \psi)$

Quiz 3

Q1.

Use natural deduction to prove that

$$\forall x \exists y (P(x) \vee Q(y)) \vdash \exists y \forall x (P(x) \vee Q(y))$$

Ans:

1. $\forall x \exists y\ (P(x) \vee Q(y))$ premise
2. $\exists y\, Q(y) \vee \neg \exists y\, Q(y)$ LEM
3. $\exists y\, Q(y)$
4. y_0 $Q(y_0)$ assumption
   - x_0 arbitrary
   - 5. $P(x_0) \vee Q(y_0)$ $\forall i$
   - 6. $(P(x) \vee Q(y_0))[x_0/x]$ same as above
7. $\forall x\, (P(x) \vee Q(y_0))$ $\forall i$
8. $\forall x\, (P(x) \vee Q(y))[y_0/y]$ same as above
9. $\exists y \forall x\, (P(x) \vee Q(y))$ $\exists i$
10. $\exists y \forall x\, (P(x) \vee Q(y))$

We will now, on the next page, use $\neg \exists y\, Q(y)$ to derive the same conclusion and then line 2 and we will let us derive $\exists y \forall x\, (P(x) \vee Q(y))$.

11. $\neg \exists y\, Q(y)$

x_0 arbitrary

12. $\exists y\, P(x_0) \vee Q(y)$ $\forall e$, premise

y_0 $P(x_0) \vee Q(y_0)$ assumption

13.

14. $P(x_0)$

15. $Q(y_0)$

16. $\exists y\, Q(y)$ $\exists i$

17. $\bot$ 11, 15

$P(x_0)$ $\bot e$

18. $P(x_0)$ $\vee e$

19. $P(x_0)$ $\exists e$ 12, 13-18

20. $P(x_0) \vee Q(y_0)$ $\vee i$

21. $(P(x) \vee Q(y_0))[x_0/x]$ same as above

22. $\forall x\, (P(x) \vee Q(y_0))$ $\forall i$

23. $\forall x\, (P(x) \vee Q(y))[y_0/y]$

same as 22

24. $\exists y\, \forall x\, (P(x) \vee Q(y))$

$\exists i$

25. $\exists y \forall x\, (P(x) \vee Q(y))$

$\vee e$ 2, 3-10, 11-24.

Q2. Consider $\phi := \psi_1 \wedge \psi_2 \wedge \psi_3$

where $\psi_1 : \forall x \exists y \; R(x,y)$

$\psi_2 : \forall x \; \neg R(x,x)$

$\psi_3 : \forall x \forall y \forall z \; (R(x,y) \wedge R(y,z) \rightarrow R(x,z))$

a) Is ϕ satisfiable?

b) Can ϕ have a finite model (one where the universe has only finitely many elements)?

Ans: a) ϕ is indeed satisfiable. Consider the universe as $\mathbb{N}$ (the set of natural numbers) and interpret $R(x,y)$ as $(x < y)$ [where $<$ is the usual "less-than" ordering among numbers].

This clearly makes ψ_1 true (we can always find a y which is bigger than any given x),

ψ_2 true (x is not bigger than itself),

and ψ_3 true (if y is bigger than x and z is bigger than y then z is clearly bigger than x).

This explanation is not required. I have done it just for your benefit.

b) ϕ has no finite models.

On the contrary suppose there is a model A with a finite universe U_A.

Let us think of the elements in this universe as nodes of a directed graph where there is an edge from a_i to a_j iff $(a_i, a_j) \in R_A$.

ψ_1 says that every node has an outgoing edge.

Since there is at least one node (universe must be non-empty), we definitely have one edge. The target node of this edge must also have an outgoing edge from there.

Since these edges can never come back (not immediately, because of ψ_2, and not even after a while, because of ψ_2 and ψ_3), the universe must have infinitely many elements.